

2025 ReviewStream

ReviewStream | 40:55

All right.

It is December 18th, 2025.

This is going to be a brief 2025 review stream going over the 72 videos of 2025.

And just to review, recap, make a kind of concentrated, distilled, final end product, taking a quick look back, seeing if any AI perplexity tools can help us get another look at the videos.

Looking at anyone's comments for who's watching live.

Seeing if we have any memories, any reflections, any trends that were arising during the year or in relationship with previous years.

And looking towards what could be in 2026.

So let's collapse all these and we'll go from the top.

Okay.

We'll just reorder things as we go.

There was, well, first just at the level of the series.

And the CODA page that I'm at is in the video description.

If you look.

Our videos, our streams are sorted into different series.

And they are reflected upon here.

So one math art stream.

35 guest streams.

That's the most general kind of one-off.

Four quarterly roundtables.

The last one coming tomorrow.

One org stream.

Two model streams.

Three math streams.

This review stream.

Ten textbook groups.

Three very long videos for the Applied Active Inference Symposium.

And 12 active inference streams with some sort of coding and various works.

So let's just jump in.

In the first.

We have Feedback Welcome with Andrew Penland.

And let's just see if Perplexity can summarize this video.

This was a fun video.

Because it involved.

In contrast with pretty much any other video.

A very specifically visual component.

Using OBS for feedback.

Kind of like that.

So that's fun to play with the format.

And talk about feedback.

Talk about strange loops.

All these different kinds of things that come up.

In the course of math.

But also get reflected on in the art.

We had three quarterly roundtables so far.

And the fourth one will be happening tomorrow.

These are organizational update videos.

Moving from the practice of organizations into the theory of organization.

Orgstream 9 with Manuel Hoffman was on the value of open source software.

And.

Here is what.

Comes from that Perplexity on Feedback Welcome.

Interesting.

This was just a generic.

But in other cases.

Perplexity is able to have some access to.

Live streams.

But appears to have some issues.

But on this one.

It.

Came out just fine.

It's all good.

We won't really.

Work with the Perplexity.

We'll just see what.

Reflections people manually have.

This org stream.

Was awesome.

I think it's very important.

To explore this question.

Of the.

The value.

And the support.
Of open source software.
And I think it.
It provides a.
A great.
Starting point.
For talking about.
How do we evaluate.
The value of.
What is.
Made.
Openly.
And.
You can.
See.
Their kind of evaluation.
Strategies.
Which.
Are.
The norms.
In economics.
And.
Talk about other.
Think about other ways to.
Value.
Software.
So.
Very cool.
To see this kind of.
Pragmatism.
Broad.
And to.
Bring value to.
What is often.
Easy to dismiss.
As.
Simply free.
In.

This year.

There were two.

Model streams.

Earlier.

In the year.

In March.

And April.

Model streams.

Are intended.

For.

Kind of technical.

Walkthroughs.

On generative models.

So.

We had.

Model stream.

17.

With.

The.

The.

Rx.

And.

Fur.

Team.

And.

With.

Frazier.

And.

This was a.

Provocative.

Presentation.

About.

What is wrong.

With active inference.

Of course.

A bit of a.

Hegelian approach.

Here.

To then.

Propose their own.

Work.

But it was a great.

Follow on discussion.

And.

Led.

Nicely.

Into.

A lot of the.

Ongoing work.

That's happening.

With.

Rx.

In.

Model stream.

18.

This was a.

Presentation.

On.

Distributantly.

Robust.

Free energy principle.

For decision making.

Okay.

I'm going to update the title.

In the.

Production document.

Just for context.

This.

This.

Coda page.

That.

I'm referencing.

Is.

Where.

All of the.

Live stream information.

Is coordinated.

The video is open to everybody.

With the link.

You won't have.

Edit access.

But that's.

Where.

The.

Protocol.

The procedure.

For how we.

Schedule.

And do.

Live streams.

And.

Who people have brought up.

To invite.

All these kinds of things.

And in this.

Video.

They.

Presented on their.

Robotics.

Research.

And about.

How to.

Deal with.

Robustness of decision making.

When there's uncertainty.

About the.

Robots.

World.

Okay.

There's.

Of course.

This review stream.

Strange loopy.

As it may be.
There were.
Three.
Math streams.
Twelve.
Thirteen.
And fourteen.
These.
Were some of.
The most.
Clear.
Memories.
For me.
I.
I.
I.
Always.
Enjoy.
These.
Mouth streams.
In.
Twelve.
One.
Hola.
Running.
Uh.
Presented.
On the.
The.
Elbowing.
Stein.
So.
Evidence.
Lower.
Bound.
Is.
Equivalent.
Essentially.

To.
Variational.
Free.
Energy.
So.
Just.
Sorry about.
That.
We should be back.
Um.
Very nice.
Statistical methods.
Covering some of these distributional update.
Rules.
And.
And ways to.
Go back and forth.
And to use these particles.
Inject these particles in.
As tracers.
On a vector field.
Mathstream.
Thirteen.
Disco.
Pie.
Monoidal categories for active inference.
Of interest.
For.
Those.
Who.
Are.
Exploring the.
Category theory.
Active inference interface.
And.
Disco.
Pie.
Is a.

Programming.

Package.

That.

That.

Helps represent.

String diagrams.

So.

Here's a string diagram.

For cooking.

And.

Alexi.

Goes into.

String diagrams.

For.

For active inference.

And.

Gives some background on.

Category theory.

Then.

In.

Mathstream.

Fourteen.

Jesse.

Von.

Ustrum.

Presented.

On.

His.

Review.

Paper.

A.

Concise.

Mathematical.

Description.

Of.

Active.

In.

Discrete.

Time.
So.
Great.
Concise.
Pedagogical.
Materials.
There were.
More than 10.
But I guess only 10.
Represented here.
Textbook groups.
This represented.
Some of the.
Closing.
Times.
Probably.
For this cycle.
At least.
That we are.
Working through.
The.
2022.
Active inference.
Free energy principle.
In.
Mind.
Brain.
And.
Behavior.
Textbook.
As.
We are.
Eagerly.
Looking towards.
The fundamentals.
Of active inference.
Principles.
Algorithms.

And.

Applications.

Of the free energy principle.

For engineers.

By Sanjeev Namjoshi.

Coming out in.

March.

Of next year.

So we're.

Getting excited for that.

Textbook group.

And looking back.

On these.

Textbook groups.

These are always.

Fun.

There's.

There's so much.

To this material.

And to.

Appreciate.

Andrew Boucher.

For facilitating.

A bunch of the textbook group meetings.

And to all the participants.

Who show up.

Active inference.

Streams.

These were.

Streams.

That.

That.

I.

Expose.

Ways to.

Show what I'm.

Working on.

Often.

In the active inference.

Or in the.

Meta research.

Or in the tool development space.

And.

The year.

Began with some.

Plain text.

Exploration.

April.

Agents.

Bring May simulations.

That was exploring.

Some different.

Agentic.

Systems.

The.

More.

LLM.

Coordination.

Based.

Open Manus.

As well as.

The.

Probabilistic.

Generative models.

With.

Rx.

Infer.

Cerebrum.

Began.

A.

Thread.

Related to.

Linguistics.

And.

Bayesian approaches.

To language.

No.

Org.

As a.

Organizational.

Agent.

Coordination.

Framework.

Generalized.

Notation.

Notation.

The following.

Three.

In.

Fourteen.

And.

This.

Was.

Moving.

Through.

This.

Domain specific.

Language.

For.

Generative models.

And.

Going from.

Plain text.

Triple play.

So.

From the configured.

File.

Into the.

Code.

That's executable.

The mathematical.

Representation.

Diagramatic.

Representation.

Natural language.
There was the.
Golden spike.
Moment.
For multi-agent.
Trajectory planning.
Where.
I.
Reverse engineered.
The example.
Of multi-agent.
Trajectory.
Into this.
Configurable format.
And then.
Forward engineered.
The configuration.
And.
We met in the middle.
Had a golden spike.
Moment.
And.
That is.
Hopefully.
If interested.
People.
Who.
Who want to see.
That kind of.
Industrial.
Grade.
Scientific integrity.
For generative models.
And the sound.
Of uncertainty.
Which was.
Adding some features.
Into the GNN package.

Like audio.

Format.

Rendering.

Directly.

Of generative models.

Not just their outputs.

Like listening.

To the configuration.

Start here.

Operational.

Metascientific.

Arts and crafts.

And generative education.

Yesterday.

Capped off.

A sort of.

Content.

Development.

Educational.

And research.

Tool.

Development.

Arc.

And.

Each of these videos.

Have their associated.

Code.

And.

So.

All open source.

Now we get to the.

Most.

Variable.

Which will be the.

Symposium.

And the guest stream.

So the.

Applied active inference.

Symposium.

The fifth.

One.

Was in.

November.

2025.

And.

Here was the.

Public program.

Just for reference.

We had three.

Three days live streams.

Keynote.

By Carl Friston.

Bunch of live streams.

Some more workshop.

Sections.

And a bunch of.

Pre-recorded.

Videos.

And.

We have the.

Timestamp information.

Available.

For all of them.

Those have been added.

To.

The videos.

So about.

Forty or fifty.

Presenters.

Really awesome.

To see.

And.

Looking forward to the next year's.

Symposium.

And then just.

Last section here.

For this.

Video.

Is.

Looking at the.

Guest streams.

And.

Let's see here.

What trends.

Might emerge.

We had.

Ritvik Praki.

Self.

Active inference.

For self-organizing.

Multi-LLM systems.

A Bayesian thermodynamic.

Approach to adaptation.

This was.

Very.

On.

Trend.

I think.

For 2025.

Looking for.

Similarities.

Differences.

Convergences.

Synergies.

With.

Active inference.

Probabilistic modeling.

Cognitive modeling.

And LLM.

Which.

Was.

One of the themes.

Of the year.

In January.

Ron Edelman.

Presented on.

Cognitive.

Experience design.

Designing AI systems.

For trust.

Also.

Complementing.

I think.

Another part of that trend.

Which is.

Not just at the.

Technical.

Back end.

Side.

Side.

But also.

In terms of the.

Human computer interface.

User interface.

Side.

How do we.

Design.

How do we.

Think about.

These.

Concepts like.

Trust.

Matt Sims.

Presented.

On.

Slime mold.

And philosophy.

And.

Matt.

Had.

Presented.

Here's the.

Full.
Video table.
Matt.
Had joined us.
In.
October.
2021.
To discuss.
His paper.
How to count.
Biological minds.
Symbiosis.
The free energy principle.
So.
Classic.
Dave Douglas.
Stephen Sillette.
Joining as well.
Uh.
And.
Joined us.
This year.
To.
Discuss.
His book.
On slime mold.
Which was a.
Book I.
Got the physical version of.
And read.
Nice.
Quick.
Concise.
Read.
In February.
We had.
The group.
Of.

Murray Smith.

Presenting.

On their work.

On human.

Computer.

Interaction.

So here.

And with.

Arun.

Who.

Who.

Coordinated.

This.

Presentation.

There's.

So.

So many.

So many.

Interesting.

Analyses.

I think.

When we look across.

All the videos.

Which.

Which visual.

Representations.

Of.

Active inference.

Of.

Of the agent.

Environments.

Do people use.

What's the rhetoric.

What's the technical.

Justification.

For the different kinds of works.

What.

What do people bring up.

As.

Their.

Their justification.

What do they highlight.

As their.

Their challenges.

And their successes.

This group.

In particular.

Put in a lot of work.

To making.

The different stages.

Of their.

Generative models.

Operation.

Very clear.

So.

Great.

Presentation.

Guest stream.

90.2.

With Benjamin Nelson.

A.

Intern.

This was.

I was in a coffee shop.

And there were some disruptions.

Which.

I guess.

Was also sort of.

A theme of the year.

And.

It was very.

Interesting.

Although.

All over the place.

And.

I hope.
Was a useful.
Milestone.
And marker.
On.
Benjamin's journey.
Going into May.
Avel.
Glennon Carleau.
Presented on.
From contextuality.
To social constraints.
An active inference.
Ontology.
For participatory realism.
And social change.
Those familiar.
With Avel's work.
Will.
Know he always brings.
A very rich.
Very holistic.
View.
To the technical.
Elements of the work.
And extends.
And connects it.
Into.
Social sciences.
And some may.
Remember.
A different version.
Of this.
Image.
In the.
2023.
Course.
On the social sciences.

Guest stream.

100.1.

Polyndrome.

With you.

Kanazawa.

On the free energy principle.

And implications.

For language learning.

And education.

This was a very.

Memorable stream.

For me.

Partially.

I think it was.

Nighttime.

Could have been.

Could have been.

Morning in Japan.

Use.

Style.

Is extremely.

Energetic.

And.

Dense.

Visual.

Information.

That really brought together.

A great number of points.

And.

Connected it to.

Language learning.

And education.

Which.

Is.

Something we've all.

Done.

And or.

Are doing.

So.

This was a.

Very.

Fun and relevant.

Presentation.

Into.

April.

A little bit of a busier.

April.

Guest stream.

101.

With.

Mohsen Jafari.

And.

Yulin Lee.

Was on.

Navigating.

Autonomous vehicles.

On unmarked road.

With the.

Fusion based.

Motion prediction.

And active inference.

So this is.

Getting into.

The applications.

In robotics.

And.

Autonomous vehicles.

And.

And.

Smart homes.

And.

So.

And.

Also.

To the more.

Technical theme.
Of similarities.
And differences.
With generative AI.
And active inference.
Here we saw.
The diffusion modeling.
And the way that.
The forward and reverse.
Diffusion modeling.
Could be used.
Which.
Came up in.
Some other streams.
4.
9.
25.
2 squared.
3 squared.
5 squared.
With.
Majeed Benny.
Who.
We will look at his.
Some of his previous.
Times joining.
Here was a.
Great stream.
With Dean.
And.
Majeed.
Always has a lot of.
Very interesting.
Philosophical points.
He joined.
Interesting.
Majeed was the first.
Guest stream.

In February.

2021.

A critical analysis.

Of Markovian monism.

Very first guest stream.

Oh.

Total throwbacks.

Total throwbacks.

And.

So.

While.

What's new.

And recent.

Is great.

It's also.

Really nice.

To.

Look.

Look back.

At some of these.

Older.

Presentations.

See where things were.

And.

The paper.

And then.

In.

Interesting.

2022.

Did.

Majeed.

Yes.

Majeed.

Worked on.

Tale of two architectures.

And then.

In the.

Of course.

Epic.

Dean infused.

Livestream 47 series.

On the double play.

With inactive.

Social.

Cognition.

And abduction.

The abduction thread.

Continued to be.

Developed by Majeed.

And colleagues.

Still in April.

This year.

We had.

82.5.

With Robert.

Warden.

A.

Institute.

Research fellow.

On computers.

Meaning.

And consciousness.

And this was part of a.

Series.

By.

Warden.

He has now.

Presented on.

Seven.

Guest dreams.

That took us.

From the.

Requirements.

Equation.

Three-dimensional.

Spatial cognition.
Projective wave.
Theory of consciousness.
And assessment of the.
Brain wave.
Hypothesis.
That was last year.
And then this year.
In April.
And October.
Was computers.
Meaning.
And consciousness.
Unified theory of language.
And language.
And human nature.
This was a.
A raucous.
Round table.
Discussion.
And.
Had.
A bunch of guests.
In.
April.
Tobias.
Tobias.
Plotman.
Presented on.
The mind in motion.
A systems model.
Of real-time.
Cognitive transitions.
So.
This was.
A.
Cognitive modeling.
Approach.

That.
Tobias.
Had.
Developed.
April.

April.
Tanahiro Taniguchi.
Joined for.
Collective predictive coding.
And active inference.
A new perspective on emergent communication.
And.
Let's.
Look at.
He also joined.
Later.
In.
Guest stream.
118.
For co-creative learning.
Via Metropolis Hastings interactions.
And.
These were both.
Excellent presentations.
Also.
Nighttime.
For me.
Also.
Fantastic.
Visually.
Comprehensive.
Technically.
Distilled.
Very nice.
Presentations.
Bringing together.
The fundamental.

The biological.

The generative.

Active inference.

And.

In.

The.

Second.

Of Tadahiro's.

Presentations.

On the channel.

He.

Talked more about.

Extending from the collective.

Predictive coding.

Into the.

Metropolis Hastings.

Naming game.

And about.

This sort of third.

Option.

That is neither supervised.

Nor unsupervised.

But co-creative.

And so that was very.

Thought provoking.

And.

Is a.

Great example of where.

Something can have both a.

Narrow technical.

Contribution to make.

As well as a.

Broader.

Philosophical point.

And.

And.

Tadahiro.

Really spanned that.

Gammon.

Into May.

Paolo.

DeMaio.

Joined.

To discuss.

Systematic.

Deviation.

And.

Presented on.

A.

A variety of.

Of.

Topics.

From.

From the chemical.

To the societal.

To the societal.

Guest.

Stream.

106.

Was with.

Bang Lu.

And.

Collaborators.

And.

They had.

Written.

This very large.

Review.

And synthesis.

Paper.

On foundation.

Agents.

And.

Presented on.

Each of the authors.

Presented on.

On different.

Sections.

Giving.

A really nice.

Almost textbook.

Like.

Presentation.

Summarizing.

The.

The state of the field.

And.

Great.

Place.

To.

To learn about.

Agentic.

And multi-agentic.

Systems.

Guest.

Stream.

107.

David.

Benermo.

Who's one of the.

IY.

Organizers.

Presented on.

Computational.

Psychiatry.

And.

Affective.

States.

Some of the modeling work.

That.

Their group.

Are involved in.

Guest.

Stream.

108.

With.

Christo K.

Thomas.

Next generation.

Artificial intelligence.

For emergent.

Semantic communication.

This.

Though.

Of course.

I don't play.

Favorites.

Was one of.

The most memorable.

Presentations.

Uh.

For me.

Also.

With.

With excellent.

Dense.

Visual.

Elements.

But.

Getting into.

This question.

Of.

Applied.

Semantic.

Transfer.

And.

Semantic.

Information.

Contents.

Which.

Had.

Been.

A.
Key theme.
In our.
Learning.
Going.
All the way.
Back.
To.
2021.
With.
Chris Fields.
Et.
All.
And.
The.
Information.
Flow.
In a.
Context.
Dependent.
Hierarchical.
Bayesian.
Inference.
And.
Christo.
Presented.
On this.
In the.
Mobile.
Networks.
Similar to.
Totahiro's.
Work.
In a way.
Regarding.
Multi-agent.
Semantic.
Transfer.

And.
Just.
Very.
Fascinating.
Discussion.
About.
How.
As our.
Communication.
Networks.
Move.
From.
Syntax.
Aware.
Bandwidth.
Like.
How many bits.
Of.
Information.
Can be transferred.
How much.
How many hard drives.
Will it fill up.
Into the semantics.
Of information transfer.
We get into this.
Real cognitive.
Modeling.
Setting.
Where we're talking.
About.
Belief.
Updating.
Message.
Passing.
And.
You could have.
One gigabyte.

File.
That.
In.
Practical.
Pragmatic.
Terms.
Doesn't.
Provide.
Any information.
And you could have.
One.
Bit.
Of.
Shannon.
Information.
That provides.
Important.
Reduction.
Of.
Uncertainty.
For a given.
Agent.
In a moment.
In a context.
So how do we really.
Work with that.
And that's where.
Active inference.
Came into play.
And so it was a great.
Discussion.
Alexander Hemming.
In.
109.
Presented on.
Free energy.
Attentional.
Narrowing.

And cognitive.

Rigidity.

This.

Was.

I think.

A.

A.

A.

Sort of.

Cousin.

Approach.

To that of Tobias.

In that this was a.

Um.

Cognitive modeling.

Mathematical.

And computational model.

That was connected to a lot of day-to-day.

Uh.

Aspects.

Cognitive systems.

Cognitive systems.

And.

Uh.

It's.

Just cool to see all these snapshots of work that's.

Ongoing.

In.

June.

Takuya Isamura.

Joined.

For triple equivalence for the emergence of biological intelligence.

And.

This.

Was.

An excellent presentation.

Takuya had joined.

Previously.

In.

2022.

In the discussions of.

Livestream 51.

On canonical.

Canonical neural networks perform active inference.

So.

Here.

Was the double play.

In 2022.

Was.

Neural network.

Loss.

Function.

And the.

Morphism.

With the variational base.

And then.

In 2025.

He.

He took.

That two.

To a three.

Went into the triple play.

And in the triple play.

Brought in.

Not just the neural network.

And the variational base.

But the Turing machine formulation.

So.

That.

Went.

That.

Another.

Fantastic.

Visually.

Exciting.

Technically.

Sophisticated.

Work.

Bringing together.

Some of the.

Major.

Continents.

Really.

In.

In the.

Intelligence.

Landscape.

Today.

The.

The neural network.

The Bayesian.

And.

The Turing.

So.

Great.

And.

Charismatic.

Fellow.

So.

Guest.

Dream.

111.1.

With Georgi.

Georgiv.

Was on.

Modeling.

And.

Predicting.

Self-organization.

And.

Dynamic.

Systems.

Out of.

Thermodynamic.

Equilibrium.

This was a.

Nice.

Long.

Presentation.

From.

Professor.

Georgiv.

And.

Involved.

A variety.

Of.

Theoretical.

And.

Empirical.

Features.

Of.

Far from.

Equilibrium.

Systems.

So.

We had a great.

Broad.

Scoping.

Discussion.

Guest.

Dream.

112.

With.

Hadiv.

Afai.

On.

Brain-like.

Variational.

Inference.

Kind of.

A classic.

Post-doc.

Stream.

Great.

Background.

We see.

The elbow.

The evidence.

Lower bound.

Negative.

Free energy.

Coming into play.

Highlighting.

The unifying.

Nature.

Of.

Variational.

Free energy.

Minimization.

Coming at this.

From more of a.

Statistical.

Families.

Of distributions.

Style.

Rather than.

For example.

The.

More.

2010.

Pristen.

Pristen.

Styled.

Unifying.

Tree.

Where here's more of.

Different.

Kinds of cognitive phenomena.

Optimal control.

Model evolution.

And selection.

Whereas.

Hadid.

Made a.

Very.

Clear.

Walkthrough.

Of.

Variational inference.

And the elbow.

And presented on.

On.

This.

Excellent.

Statistical work.

113.

With.

Sergio.

Yakes.

On.

Bayesian mechanics.

Of economic choice.

Computational foundations.

Of economic behavior.

This was a.

Great stream.

Brought together.

A lot of the.

History.

And classics.

And fundamentals.

Of micro.

And macroeconomics.

And got all the way.

On through.

The.

Kinds of modeling.

That Sergio.

And colleagues.
Are engaged in right now.
And I know has developed.
Significantly.
Since this.
Presentation.
As well.
Guest stream.
114.
Chris Fields.
Adam Saffron.
Maxwell Ramsted.
And Carl Friston.
Was on the.
Inner screens.
Paper.
So the inner screen.
Paper.
Related to.
Nested blankets.
To quantum active.
Inference.
And reference frames.
And to awareness.
And consciousness.
So.
As one could imagine.
It brought out a lot of.
Fun.
Thought provoking.
Ideas.
I remember.
This.
Late July.
Early.
August.
Period.
Where.

There was kind of a.
A turn to energy.
And.
Alexei Gladstone.
Presented on.
Energy based.
Transformers.
Are scalable.
Learners.
And thinkers.
Transformers.
And this was kind of like.
Bringing together.
Energy based models.
And transformers.
And was a very.
Sparse.
Concise.
Clear.
Relevant.
Presentation.
That is.
Definitely.
Definitely.
Important.
For those who are.
Looking at.
Energy based models.
And about.
What.
Where.
Where transformers.
Can.
Can go from here.
Then taking a little bit more.
Of a.
Physical.
Material.

Perspective.

Max.

Safer.

Presented on.

Solving the compute.

Crisis.

With physics based.

ASICs.

Those are application specific.

Integrated computers.

And.

Max.

Was first author.

On this.

Review and perspective piece.

So basically.

He.

Took us through.

The.

Paper.

That he wrote.

With colleagues.

And.

For.

Those interested.

In.

Unconventional.

And.

And.

Conventional.

Computing.

It was a.

A great.

Paper.

And.

Obviously.

With.

Critical.

Economic.

And.

Pragmatic.

Relevance.

Going into.

Guest stream.

117.

Robert.

Stamps.

Presented on.

Active inference.

And artificial spin.

Ice.

Control processes.

And state selection.

Here's.

Vladimir.

And.

Robert.

Was presenting.

On the.

Nanomagnet.

So.

Material.

Substrate.

And.

How predictive coding.

Was being implemented.

On this material.

Nanomagnetic.

Substrate.

118.

Was.

Todd.

Hero's.

Second.

Talk.

Guest stream.

119.

With.

Nick.

Chater.

And.

Robert.

McKay.

This was.

Another.

Economics.

Presentation.

And.

This one.

Was.

A little bit.

More.

Coming from.

The macroeconomic.

Side.

Side.

Still.

Very.

Market.

Oriented.

And.

Some.

Exciting.

Developments.

In terms of.

Understanding.

Markets.

From first principles.

Guest stream.

120.

Was with.

Ifidio.

Manuel.

And.

He.
Presented.
On.
Several.
Projects.
Related to.
Temporality.
Of.
Cognitive.
Systems.
And.
Physics.
Of.
Of.
Eye.
Movement.
Brought in.
Some.
Consciousness.
Topics.
And.
So.
It was.
Fun.
And.
Interesting.
Talk.
Getting.
To the last.
Two.
Here.
Guest stream.
121.
Mike.
Wiest.
Conscious.
Active.
Inference.

This.

Brought together.

Some.

With.

With.

With.

A very provocative.

Cover slide.

Might I add.

On.

Quantum.

Active.

Inference.

Less.

From.

The.

Information.

Processing.

And.

More.

From.

The.

Material.

Substrate.

Of.

Quantum.

Orc.

Or.

That type.

Of.

Area.

Connected.

To.

The.

Cognitive.

And.

Also.

The.

Consciousness.

Aspects.

Warden.

Guest.

Stream.

122.

Was.

With.

Andrea.

Hilt.

And.

She.

Presented.

On.

Wayfinding.

Radical.

Relations.

Scaling.

Without.

Hierarchies.

Models.

Without.

The.

Meat.

Great.

Neuroscience.

Informed.

Philosophy.

And.

And.

And.

Always.

A.

Good.

Experience.

To listen.

And.

Learn.

From.

How.

She's.

Thinking.

Through.

These.

Things.

Warden.

Again.

On.

11.5.

Was.

Marcelo.

Guzman.

Who.

Presented.

On.

Learning.

In.

Physical.

Networks.

So.

This.

One.

I would.

Say.

Was.

A.

Great.

Cousin.

Of.

Robert.

Stamps.

In.

Terms.

The.

Nanomagnetics.

And.

Here's.

Another.

Example.

Of.

Using.

Physical.

Unconventional.

Computing.

Substrates.

And.

Big.

Discussion.

Topics.

Around.

Well.

Okay.

What are.

The areas.

That they can.

Hot.

Swap.

In.

What are.

Kinds of.

Cognitive.

Or.

Computational.

Functions.

That.

These.

Unconventional.

Substrates.

Can.

Immediately.

Be.

Used.

For.

And.

How.
Do.
We.
Program.
How.
Do.
We.
Work.
With.
Them.
How.
Do.
We.
Model.
Them.
Guestream.
124.
Ty.
Roachford.
Who.
Presented.
On.
PCT.
And.
FEP.
And.
Discuss.
Some.
Of.
His.
Work.
In.
Comparing.
And.
Holding.
Up.
Active.
Inference.

Alongside.
The.
Perceptual.
Control.
Theory.
Kinds.
Of.
Models.
And.
Connecting.
Them.
With.
Category.
Theory.
And.
Even.
Just.
Yesterday.
Martin.
Nika.
And.
Osman.
Presenting.
On.
From.
Charles.
Darwin's.
Root.
Brain.
To.
Nikola Tesla's.
6G.
World.
Brain.
And.
X.
AI.
Native.

6G.

Networks.

Similar.

To.

Christo.

Thomas's.

Presentation.

In.

The.

Focus.

On.

Telecommunication.

Networks.

With.

A few.

Different.

Flavors.

Brought.

In.

That.

Was.

Very.

Interesting.

And.

A.

Great.

Power.

Outage.

Disruption.

So.

If.

Anyone.

Has.

A.

Comment.

Write.

It.

Otherwise.

That.
Will.
Basically.
End.
It.
Just.
Wanted.
To.
Touch.
Upon.
Each.
Of.
Them.
Show.
One.
Last.
Kind of.
Distilled.
Artifact.
Of.
What.
Happened.
This.
Year.
And.
Look.
Towards.
Next.
Year.
So.
If.
You.
Have.
An.
Idea.
Get.
In.
Contact.

Leave.

A.

Comment.

And.

Hope.

These.

Were.

Epistemic.

And.

Or.

Pragmatic.

For.

You.

Thanks.

For.

Watching.

For.

Participating.

Along.

And.

See.

Next.

Time.

Next.

Thank you.